Parallel computing of Virtual Element Method with new mesh tools

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1 Laplace equation

Let consider the Laplace equation:

$$\left\{ \begin{array}{rcll} -\Delta \boldsymbol{u} & = & \lambda \boldsymbol{u} & \text{in} \quad \Omega \\ \boldsymbol{u} & = & 0 & \text{on} \quad \partial\Omega \end{array} \right. \tag{1}$$

and $\boldsymbol{\mathcal{V}} = H_0^1(\Omega)$, the associated functional space. The variational formulation of (1) is given as follow:

Find $(\lambda, \boldsymbol{u}) \in \mathbb{R} \times \boldsymbol{\mathcal{V}}$ **with** $\boldsymbol{u} \neq 0$ **such that:**

$$a(\boldsymbol{u}, \boldsymbol{v}) = \lambda\, b(\boldsymbol{u}, \boldsymbol{v}) \qquad \forall \boldsymbol{v} \in \boldsymbol{\mathcal{V}}$$

where

$$a(\boldsymbol{u}, \boldsymbol{v}) = \int_\Omega \nabla \boldsymbol{u}\, \nabla \boldsymbol{v} d\boldsymbol{x} \quad \text{et} \quad b(\boldsymbol{u}, \boldsymbol{v}) = \int_\Omega \boldsymbol{u}\, \boldsymbol{v} d\boldsymbol{x} \qquad \forall \boldsymbol{u}, \boldsymbol{v} \in \boldsymbol{\mathcal{V}}.$$

Let us also consider the discrete functional space $\boldsymbol{\mathcal{V}}_h^E \subseteq \boldsymbol{\mathcal{V}}$ such that:

$$\boldsymbol{\mathcal{V}}_h^E := \{\boldsymbol{v}_h \in H_0^1(E) \colon\ \boldsymbol{v}_h|_{\partial E} \in \mathcal{C}^0(\partial E),\ \boldsymbol{v}_h|_e \in \boldsymbol{\mathcal{P}}_k(e)\ \forall e \in \partial E,\ \Delta \boldsymbol{v}_h \in \boldsymbol{\mathcal{P}}_{k-2}(E)\}.$$

The subspace that we will consider for the projection operator is $\widehat{\boldsymbol{\mathcal{V}}}_h^E = \boldsymbol{\mathcal{P}}_k(E)$ and the projection operators:

$$\Pi_a^E, \Pi_b^E \colon \boldsymbol{\mathcal{V}}_h^E \longrightarrow \widehat{\boldsymbol{\mathcal{V}}}_h^E$$

associated to $a(\cdot, \cdot)$ and $b(\cdot, \cdot)$, respectively.

VEM formulation

We have, for all $\boldsymbol{u}_h, \boldsymbol{v}_h \in \boldsymbol{\mathcal{V}}_h^E$,

- **local rigidity term:**

$$a_h^E(\boldsymbol{u}_h, \boldsymbol{v}_h) = a_h^E(\Pi_a^E \boldsymbol{u}_h, \Pi_a^E \boldsymbol{v}_h) + S_1^E(\boldsymbol{u}_h - \Pi_a^E \boldsymbol{u}_h, \boldsymbol{v}_h - \Pi_a^E \boldsymbol{v}_h);$$

- **local mass term:**

$$b_h^E(\boldsymbol{u}_h, \boldsymbol{v}_h) = b_h^E(\Pi_b^E \boldsymbol{u}_h, \Pi_b^E \boldsymbol{v}_h) + S_2^E(\boldsymbol{u}_h - \Pi_b^E \boldsymbol{u}_h, \boldsymbol{v}_h - \Pi_b^E \boldsymbol{v}_h)$$

where $a_h^E(\Pi_a^E \boldsymbol{u}_h, \Pi_a^E \boldsymbol{v}_h), S_1^E(\cdot,\cdot)$ are respectively the consistency and the stability terms for the local rigidity and $b_h^E(\Pi_b^E \boldsymbol{u}_h, \Pi_b^E \boldsymbol{v}_h), S_2^E(\cdot,\cdot)$ respectively the consistency and the stability terms for the local mass.
For both terms, we have a consistency and a stability part. Let consider the basis subspace (for $k=1$) $\boldsymbol{\mathcal{M}}_1^E \subseteq \widehat{\boldsymbol{\mathcal{V}}}_h^E$ such that

$$\boldsymbol{\mathcal{M}}_1^E = \Big\{ \boldsymbol{m}_1(\boldsymbol{x}) = 1, \quad \boldsymbol{m}_2(\boldsymbol{x}) = \frac{x - x_E}{h_E} \quad \text{et} \quad \boldsymbol{m}_3(\boldsymbol{x}) = \frac{y - y_E}{h_E} \Big\}.$$

2 Orthotropic elasticity equation

Let consider the orthotropic elasticity equation:

$$\left\{ \begin{array}{rcll} -\mathrm{div}\boldsymbol{\sigma}(\boldsymbol{u})(\boldsymbol{x}) & = & \boldsymbol{f}(\boldsymbol{x}) & \text{in} \quad \Omega \\ \boldsymbol{u} & = & 0 & \text{on} \quad \partial\Omega \end{array} \right. \tag{2}$$

where $\boldsymbol{f} = (\boldsymbol{f}_1, \boldsymbol{f}_2)^\top \in [\mathbf{L}^2(\Omega)]^2$ and the stress tensor $\boldsymbol{\sigma}$ such that

$$\boldsymbol{\sigma} = C\boldsymbol{\varepsilon}. \tag{3}$$

with $\boldsymbol{\varepsilon}(\cdot) = \frac{1}{2}(\nabla \cdot + \nabla^\top \cdot)$ In the case of plane deformation (*Hooke's law*) of an orthotropic material, $C \in \mathbb{R}^{3\times 3}$ such that (3):

$$C = \begin{bmatrix} C_{11} & C_{12} & 0 \\ & C_{22} & 0 \\ \text{sym.} & & C_{66} \end{bmatrix} = \begin{bmatrix} \frac{E_1}{\Delta} & \frac{\nu_{21}E_1}{\Delta} & 0 \\ \frac{\nu_{12}E_2}{\Delta} & \frac{E_2}{\Delta} & 0 \\ 0 & 0 & G_{12} \end{bmatrix}.$$

C is symmetric because $\dfrac{\nu_{12}}{E_1} = \dfrac{\nu_{21}}{E_2}$.

The variational formulation associated to (2) is given by:

Find $\boldsymbol{u} \in \boldsymbol{\mathcal{V}} := [\mathbf{H}_0^1(\Omega)]^2$ such that,

$$a(\boldsymbol{u}, \boldsymbol{v}) = l(\boldsymbol{v}) \qquad \forall \boldsymbol{v} \in \boldsymbol{\mathcal{V}} \tag{4}$$

with

$$a(\boldsymbol{u}, \boldsymbol{v}) = \int_\Omega C\boldsymbol{\varepsilon}(\boldsymbol{u}) : \boldsymbol{\varepsilon}(\boldsymbol{v}) d\boldsymbol{x} \qquad \text{and} \qquad l(\boldsymbol{v}) = \int_\Omega \boldsymbol{f}(\boldsymbol{x}) \boldsymbol{v}(\boldsymbol{x}) d\boldsymbol{x} \qquad \forall \boldsymbol{u}, \boldsymbol{v} \in \boldsymbol{\mathcal{V}}.$$

VEM Formulation

For all $\boldsymbol{\varphi}_i, \boldsymbol{\varphi}_j \in \boldsymbol{\mathcal{V}}_h^E \quad i,j = 1,2,\cdots,2n_d$, we have

$$a(\boldsymbol{\varphi}_i, \boldsymbol{\varphi}_j) = \sum_{E\in\mathcal{T}_h} a_h^E(\boldsymbol{\varphi}_i, \boldsymbol{\varphi}_j) = \sum_{E\in\mathcal{T}_h} \int_E C\boldsymbol{\varepsilon}(\boldsymbol{\varphi}_i) : \boldsymbol{\varepsilon}(\boldsymbol{\varphi}_j) d\boldsymbol{x} \qquad \forall \boldsymbol{u}, \boldsymbol{\varphi}_j \in \boldsymbol{\mathcal{V}}_h^E$$

and

$$l(\boldsymbol{\varphi}_j) = \sum_{E\in\mathcal{T}_h} l_h^E(\boldsymbol{\varphi}_j) = \sum_{E\in\mathcal{T}_h} \int_E \boldsymbol{f}(\boldsymbol{x})\boldsymbol{\varphi}_j(\boldsymbol{x}) d\boldsymbol{x} \qquad \forall \boldsymbol{\varphi}_j \in \boldsymbol{\mathcal{V}}_h^E.$$

Then, for all $\boldsymbol{\varphi}_i, \boldsymbol{\varphi}_j \in \boldsymbol{\mathcal{V}}_h^E$, the associated VEM formulation is given by:

- **local rigidity term:**

$$a_h^E(\boldsymbol{\varphi}_i, \boldsymbol{\varphi}_j) = a_h^E(\Pi^E\boldsymbol{\varphi}_i, \Pi^E\boldsymbol{\varphi}_j) + S^E(\boldsymbol{\varphi}_i - \Pi^E\boldsymbol{\varphi}_i, \boldsymbol{\varphi}_j - \Pi^E\boldsymbol{\varphi}_j)$$

- **source term:**

$$l_h^E(\boldsymbol{\varphi}_j) = \int_E \boldsymbol{f}_h(\boldsymbol{x})\boldsymbol{\varphi}_j(\boldsymbol{x}) d\boldsymbol{x} = \int_E \Pi_{\overline{k}}^E \boldsymbol{f}\boldsymbol{\varphi}_j d\boldsymbol{x} \quad \text{for all} \quad \overline{k} = \max\{k-2, 0\}$$

where $\Pi_{\overline{k}}^E$ is defined as $\quad \Pi_{\overline{k}}^E = \Pi^E \quad \forall k \geq 2.$

Let consider the basis subspace (for $k=1$) $\boldsymbol{\mathcal{M}}_1^E \subseteq \widehat{\boldsymbol{\mathcal{V}}}_h^E$ such that

$$\boldsymbol{\mathcal{M}}_1^E = \{\boldsymbol{m}_\alpha(\boldsymbol{x}), \quad 1 \leq \alpha = 1,2,\cdots,6\}$$

with

$$\boldsymbol{m}_1(\boldsymbol{x}) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \boldsymbol{m}_2(\boldsymbol{x}) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad \boldsymbol{m}_3(\boldsymbol{x}) = \begin{pmatrix} -\frac{y-y_E}{h_E} \\ \frac{x-x_E}{h_E} \end{pmatrix},$$

$$\boldsymbol{m}_4(\boldsymbol{x}) = \begin{pmatrix} \frac{y-y_E}{h_E} \\ \frac{x-x_E}{h_E} \end{pmatrix}, \quad \boldsymbol{m}_5(\boldsymbol{x}) = \begin{pmatrix} \frac{x-x_E}{h_E} \\ 0 \end{pmatrix}, \qquad \boldsymbol{m}_6(\boldsymbol{x}) = \begin{pmatrix} 0 \\ \frac{y-y_E}{h_E} \end{pmatrix}.$$

Remark:

For more details on the resolution and the choice of the degree of freedom, please refer to internship documents.

3 Domain mesh tools

Let take a look at an example of a mesh (fig. 1) and all informations around it.

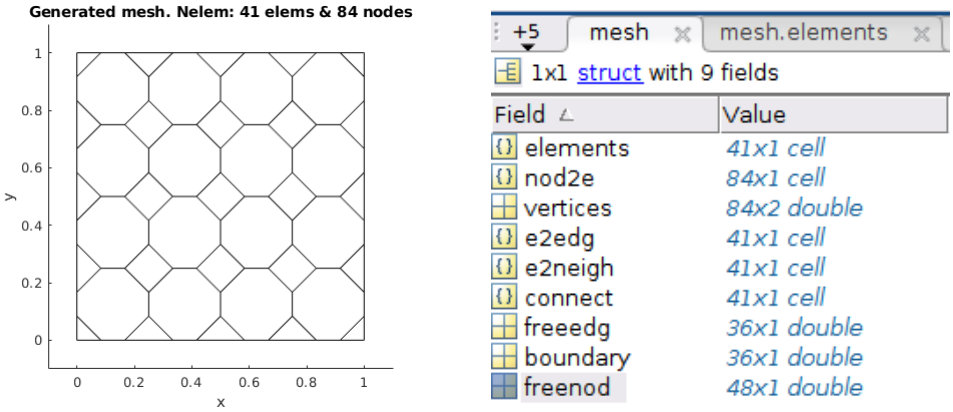


Figure 1: An example of mesh and its informations

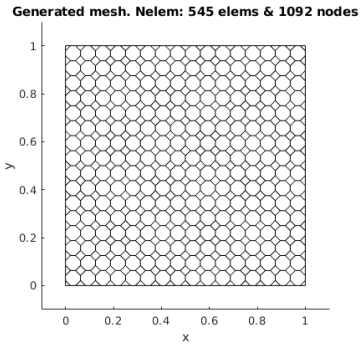
Fields	Summary
elements	a cell array that contains the node numbering for each elements
nod2e	a cell array that contains the belonging element for each nodes
vertices	a matrix that contains the (x,y) coordinates for each nodes
e2edg	a cell array that contains the edge numbering for each elements
e2neigh	a cell array that contains the neighbor numbering of each elements
connect	a cell array that contains the connectics (degree of freedom) for each elements
freeedg	an array that contains the edge numbering on the boundary
boundary	an array that contains the node numbering on the boundary
freenod	an array that contains the interior numbering nodes

Remark:

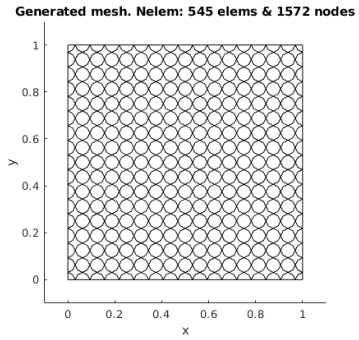
This mesh is just an example and user can also define its own geometrical form by giving the cartesian coordinates of a single element to the mesh tool (see also the Annex for other examples).

4 Numerical results

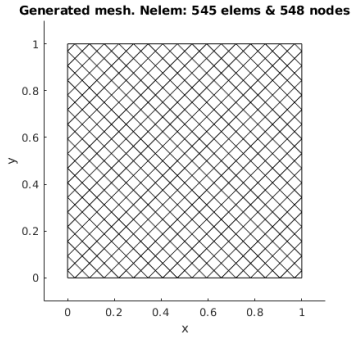
Let consider the following meshes (fig. 2) for the resolution of the problem



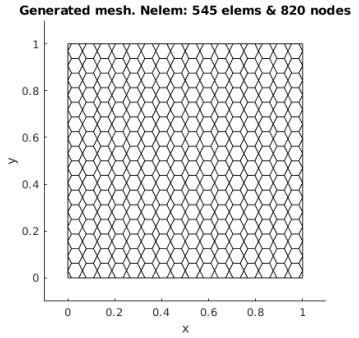
(a) mesh1



(b) mesh2



(c) mesh3



(d) mesh4

Figure 2: Meshes for the resolution

4.1 Laplace equation

Sequential resolution

Let us consider the Laplace equation

$$\left\{ \begin{array}{rcll} -\Delta \boldsymbol{u} + \boldsymbol{u} & = & (2\pi^2 + 1)\,\sin(\pi x)\sin(\pi y) & \text{in} \quad \Omega \\ \boldsymbol{u} & = & 0 & \text{on} \quad \partial\Omega \end{array} \right.$$

And different norms ($Err_{\mathbf{L}^2}$ and $Err_{\mathbf{H}^1}$) for the validation.

$$\boldsymbol{u}_{ex}(x, y) = \sin(\pi x)\sin(\pi y).$$

Parallel resolution

The resolution is made on google cloud platform and the following values are obtained:

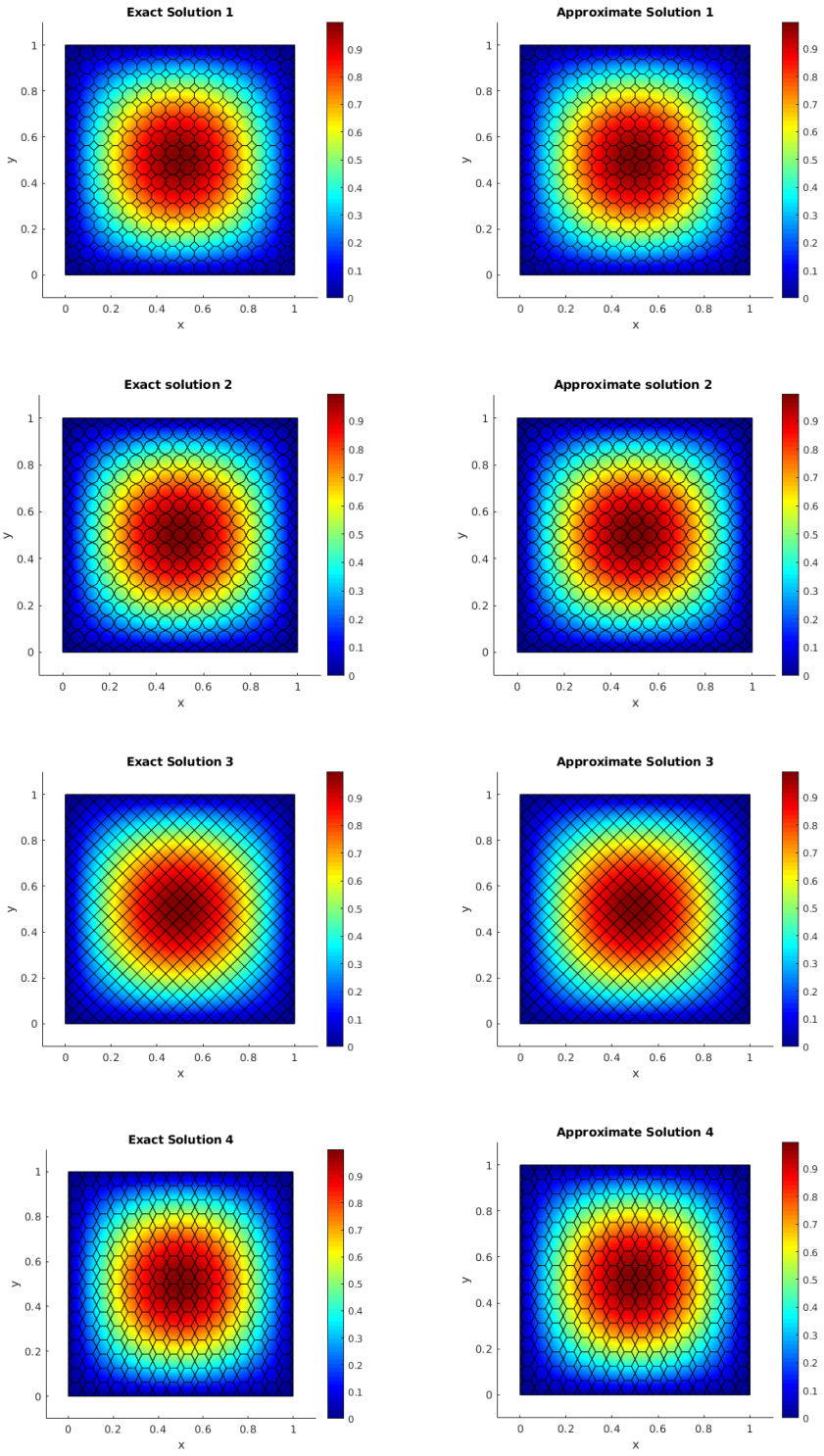


Figure 3: Resolution of Laplace equation with different meshes.

Domain	np	Assembling (sec)	Solving (sec)	$Err_{\mathbf{L}^2}$	$Err_{\mathbf{H}^1}$	Residual	last iter
	1	70,39	0,37				
mesh1	2	8,72	0,11	6,4e-4	2,9e-3	5,19e-16	17
	4	1,25	0,04				
	8	0,24	0,01				
	1	305,66	1,13				
mesh2	2	39,63	0,31	2,5e-3	5,7e-2	6,42e-14	26
	4	5,72	0,11				
	8	0,86	0,08				
	1	7,75	0,08				
mesh3	1	1,02	0,02	6,4e-4	2,9e-3	1,06e-16	14
	4	0,22	0,01				
	8	0,11	0,001				
	1	38,96	0,22				
mesh4	2	4,75	0,06	8,6e-4	1,1e-2	1,05e-13	18
	4	0,65	0,03				
	8	0,12	0,05				

An illustration (for np = 2) is given on fig. 4

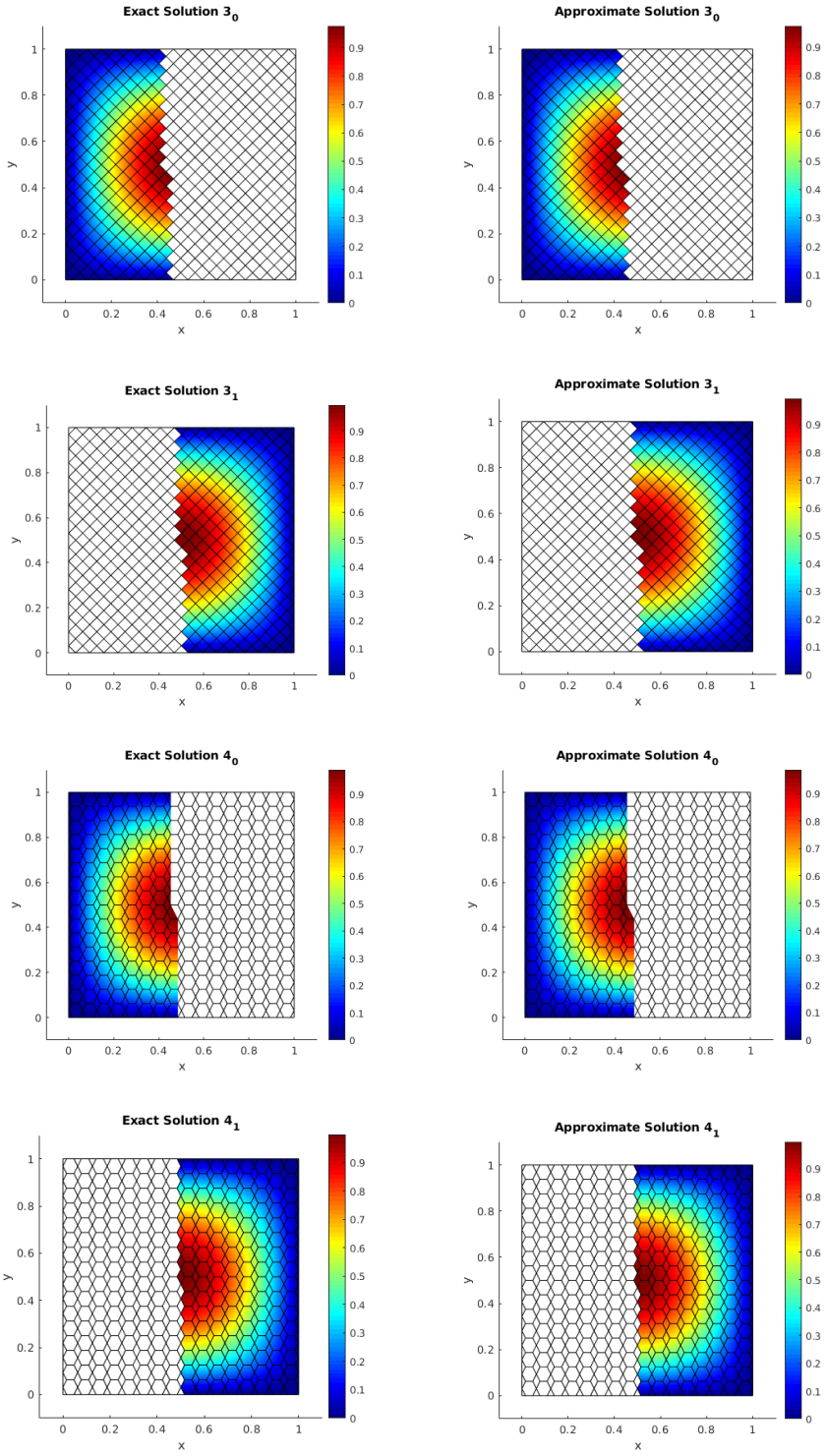
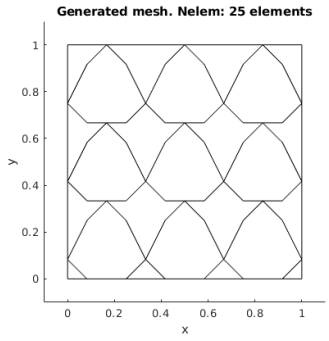
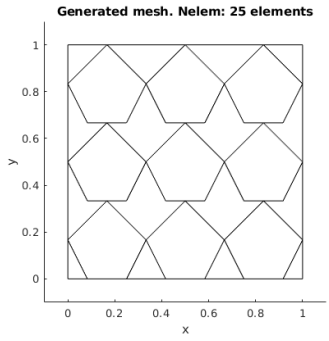
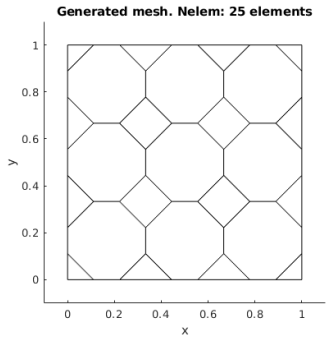


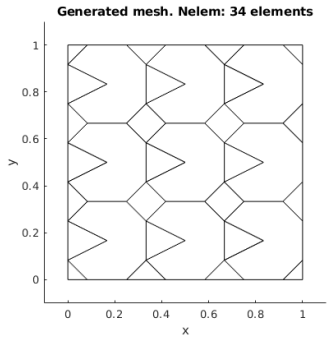
Figure 4: Parallel resolution of Laplace equation with 2 cores.

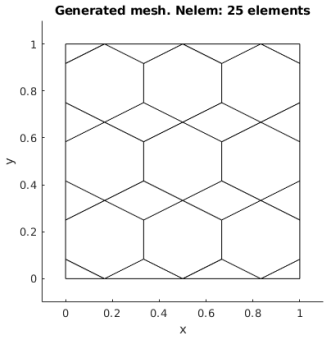
5 Annex: 2D meshes available (users can also define his own)

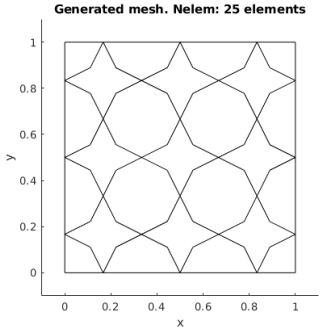


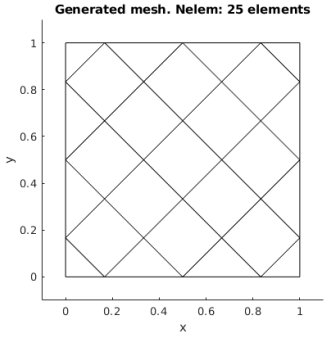


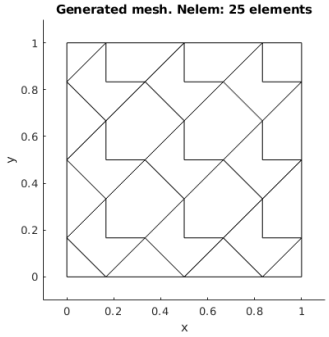


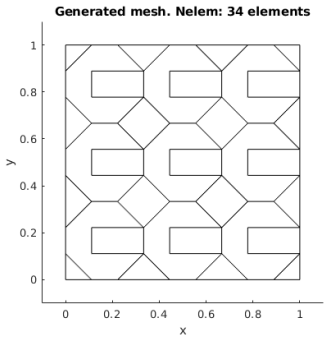


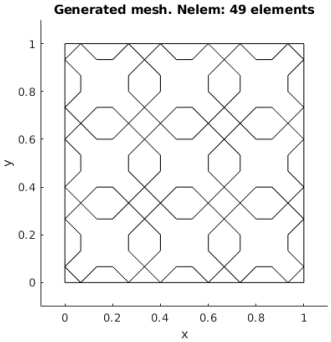

Generated mesh. Nelem: 25 elements
y
x
0
0.2
0.4
0.6
0.8
1

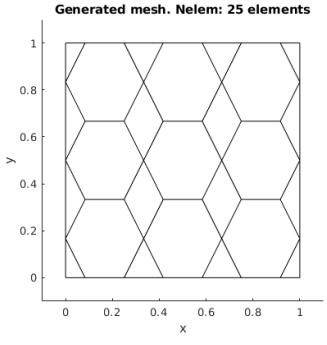

Generated mesh. Nelem: 25 elements
y
x
0
0.2
0.4
0.6
0.8
1

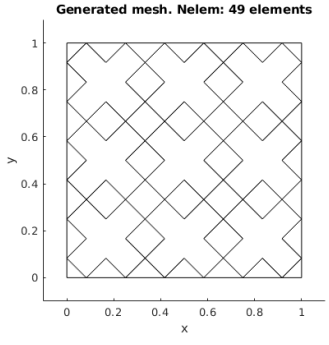

Generated mesh. Nelem: 25 elements
y
x
0
0.2
0.4
0.6
0.8
1


Generated mesh. Nelem: 25 elements
y
x
0
0.2
0.4
0.6
0.8
1


Generated mesh. Nelem: 34 elements
y
x
0
0.2
0.4
0.6
0.8
1


Generated mesh. Nelem: 49 elements
y
x
0
0.2
0.4
0.6
0.8
1


Generated mesh. Nelem: 25 elements
y
x
0
0.2
0.4
0.6
0.8
1


Generated mesh. Nelem: 49 elements
y
x
0
0.2
0.4
0.6
0.8
1

Remark:

- Users can also define his own principal element (face), report the coordinates into the algorithm, give the mesh domain ([a,b]x[c,d]) and the desire number of principal elements. Then the algorithm will charge to generate the desired mesh in taking into account the other edges and form supplementary elements. And after, it remove all unused nodes and its corresponding vertices and finally reorganize all node numbers.
- The algorithm will also give all informations about neighbours of each elements, nodes, edges, edge's connection with each elements (normal vectors) and nodes on the boundary.
- All the informations are saved in a **.mat** file that can be used.

6 3D meshes available (users can also define his own)

To be continued ...

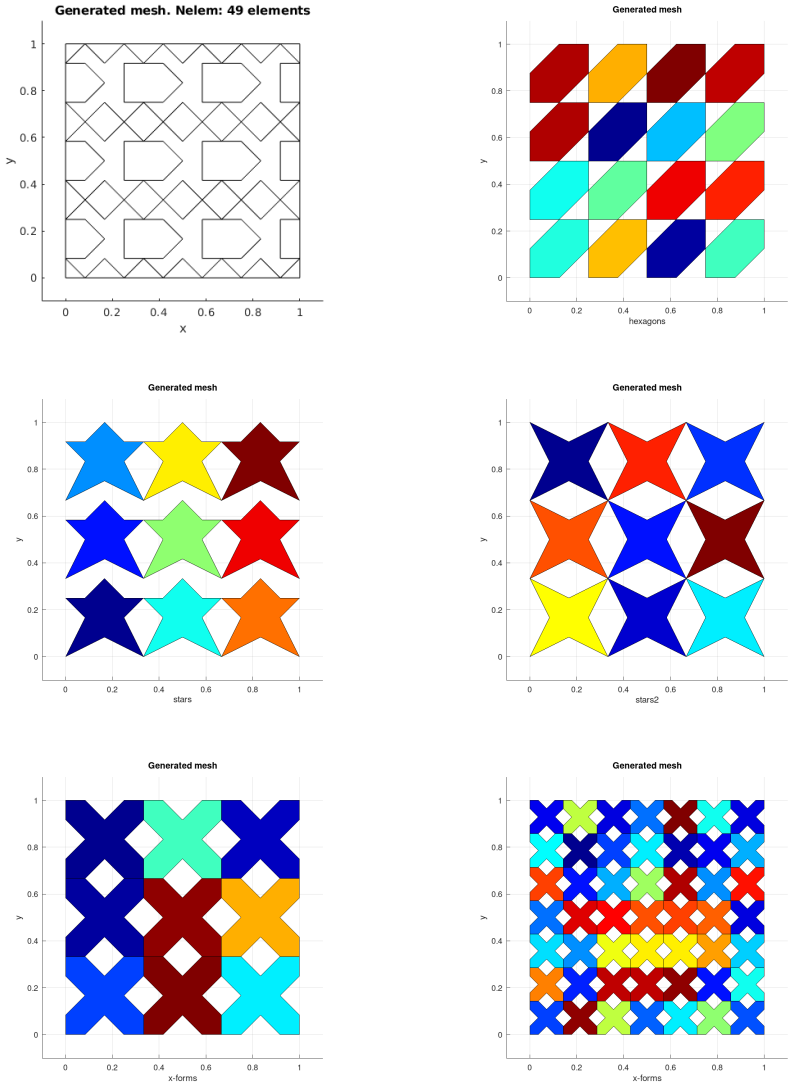


Figure 7: X-forms

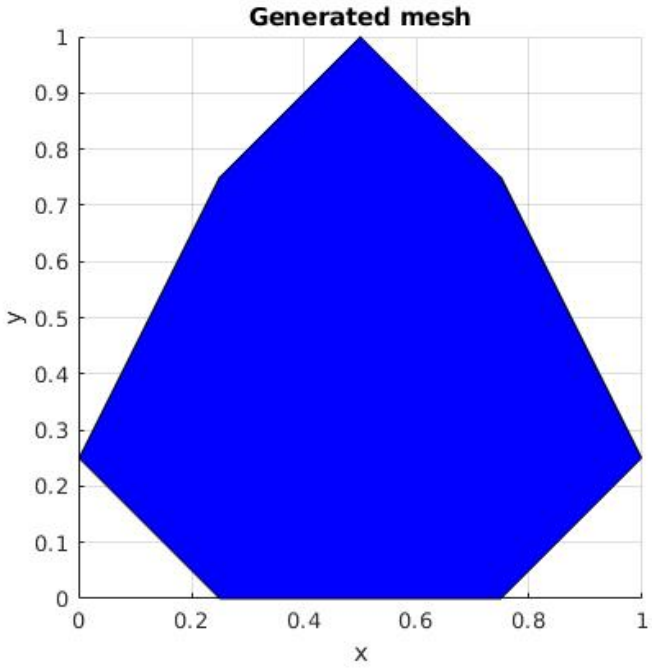


Figure 8: Example of principal element of the mesh.